MPS2L: Mutual Prediction Self-Supervised Learning for Remote Sensing Image Change Detection

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***Abstract*— In this article, we propose a novel mutual prediction self-supervised learning (MPS2L) method for remote sensing (RS) image change detection (CD). Compared with the previous self-supervised CD methods based on contrastive learning (CL), MPS2L employing a pixel-level training strategy based on masked image modeling (MIM) can effectively train the model to interpret the local scene of RS images. Utilizing global and local scenes and temporal change features extracted from masked bitemporal images to achieve cross-temporal mutual prediction makes the model have the ability to understand the overall observation scene and capture the change information. The training of the two abilities is carried out simultaneously, avoiding the problem of multiobjective conflict or mutual inhibition. To better focus on the changing regions in RS scenes, we further introduce a change feature interaction module (CFIM), comprising spatial and channel feature interaction. The channel interaction module (CIM) can facilitate the cross-temporal transmission of global scene information by channel attention, and the spatial interaction module (SIM) can promote the network to capture information on changing regions by spatial attention. The experimental results on three benchmark RS CD datasets demonstrate the effectiveness and priority of our proposed MPS2L compared to some existing state-of-the-art (SOTA) methods. The source code of the proposed MPS2L will be made available publicly at https://github.com/KustTeamWQW/MPS2L.**

***Index Terms*— Attention mechanism, masked image modeling (MIM), remote sensing (RS) image change detection (CD), self-supervised learning.**

I. Introduction

CHANGE detection (CD), a prominent area within remote sensing (RS) [1], [2], involves the systematic analysis of surface alterations occurring between distinct temporal phases within the same geographical region [3]. This process holds significant importance across various domains, including environmental assessment [4], monitoring of geological hazards [5], land cover mapping, and urban development planning [6], [7]. Within the realm of CD, deep learning (DL)-based models predominantly encompass convolutional neural network (CNN) and transformer architectures, renowned for their adeptness in feature extraction. However, underpinning the commendable performance of such methodologies lays a reliance on extensively annotated datasets.

Self-supervised learning represents a novel paradigm within DL, enabling models to extract the global and general features of images through pretext tasks, thus reducing the burden on labeled samples. Following self-supervised training, a small number of labeled images may subsequently be utilized for transfer learning on specific downstream tasks, thereby enhancing model performance. Self-supervised learning methods are principally divided into contrastive learning (CL) [8], [9], [10] and masked image modeling (MIM) [11], [12], [13]. At present, self-supervised CD methods are mainly improved on the basis of CL method. Various methods enhance detection capabilities by integrating additional pretext tasks specific to CD [14], [15] or by implementing a staged process combining CL with CD [16], [17]. Nonetheless, these methodologies encounter two primary challenges: 1) the simultaneous optimization of dual objectives might obscure critical focal points for the model, leading to ineffective or insufficient training and 2) the classification mechanism inherent in CL strategies tends to prioritize global scene feature extraction, whereas the region of change in RS images is usually local. Consequently, the model's ability to perceive and interpret change information following CL often remains inadequate.

In contrast to CL, MIM offers a pixel-level training strategy that more effectively enhances the capability of the model to discern local scenes. Furthermore, this method utilizes unmasked image patch information to reconstruct the full image, thereby enabling comprehensive scene perception. Nonetheless, in order to perform the CD task better, the model also needs to have the ability to capture the change information between the bitemporal images. Since the MIM method reconstructs only a single masked image at a time, it falls short in training the model to recognize temporal changes. We contend

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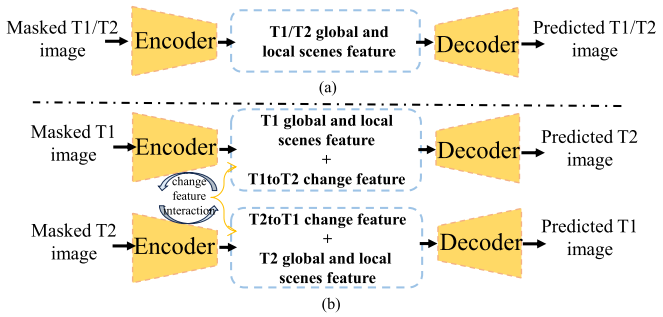


Fig. 1. (a) MIM. (b) Cross-temporal mutual prediction. The MIM extracts the global and local scenes features of the masked image through the encoder and inputs them to the decoder to realize the reconstruction of the original image. Mutual prediction extracts the global and local scenes and temporal change features of masked images through encoder and change feature interaction, respectively, and inputs them to the decoder to achieve cross-temporal prediction.

that reconstructing an image from partial patch information essentially is a prediction process. Bitemporal images exhibit a high degree of overall similarity, with differences primarily attributed to change information. By supplementing the change information on the basis of global and local scene features extracted from the unmasked patches of the masked bitemporal images, the cross-temporal mutual prediction can be realized theoretically, as shown in Fig. 1. Drawing inspiration from Fang et al.'s [18] Changer, we find that the additional design feature interaction module can independently capture the change information in the encoder and complete the change information extraction.

Based on this, we propose a novel mutual prediction self-supervised learning (MPS2L) method, accompanied by a change feature interaction module (CFIM). Compared with the random interaction of features, CFIM using an attention mechanism can pay more attention to the change regions. The CFIM comprises spatial and channel interaction modules (CIMs). The spatial interaction module (SIM) obtains attention maps through shallow features and interacts with local change information. The CIM obtains the attention vector through the deep features and interacts with the global change information. The MPS2L method unfolds in two stages: self-reconstruction and mutual prediction and utilizing a twin encoder–decoder structure enhanced with CFIM. The self-reconstruction stage solely harnesses scene features extracted from partial patches to reconstruct the complete original image, thereby training the model to interpret both global and local scenes within RS imagery. In the mutual prediction stage, the encoder and CFIM extract global and local scenes and temporal change features from masked bitemporal images, respectively, for input into the decoder to accomplish mutual prediction between bitemporal images, primarily focusing on training the capability of the model to detect changes. The pixel-level MIM strategy ensures comprehension of local scenes, and the capture of change information in the mutual prediction stage is subordinate to the prediction task, which is the only goal, so as to avoid the existence of multiple optimization objectives leading to the model not being effectively or adequately trained. By preserving the weights of the encoder and CFIM during self-supervised training, substituting a decoder tailored for CD, and employing a small number of supervised samples for transfer training on downstream tasks, the model can achieve exceptional performance.

The contributions of this article can be summarized as follows.

1) We propose a self-supervised CD method called MPS2L. MPS2L utilizes MIM to enhance the model's comprehension of both global and local RS scenes, and the mutual prediction mechanism ensures the capability to detect change regions effectively.
2) We propose a module called CFIM that can facilitate the capture of change information. CFIM obtains attention map through shallow features, interacting with local change information, and obtains attention vector through deep features, interacting with global change information. It improves the ability of the model to perceive change.

The remainder of this article is structured as follows. Section II provides a brief review of related works. Section III details our proposed method. Section IV reports experimental results. The discussion is given in Section V, and conclusion is drawn in Section VI.

II. Related Works

A. DL-Based RS Image CD

Nowadays, due to the robust feature extraction capabilities, the newly developed DL-based CD methods have demonstrated outstanding performance. Notably, Zhan et al. [19] pioneered the integration of the Siamese convolutional network into CD tasks, enabling parallel processing of bitemporal images. In addition, Daudt et al. [20] proposed three effective CD architectures based on fully convolutional neural networks (FCNNs), namely, fully convolutional early fusion (FC-EF), fully convolutional Siamese concatenation (FC-Siam-conc), and fully convolutional Siamese difference (FC-Siam-diff). Similarly, CDNet [21] employs multisensor fusion SLAM and a rapid dense 3-D reconstruction pipeline based on the Siamese architecture for pixel-by-pixel CD. These methods, along with adaptations from researchers such as STANet [22], DASNet [23], triplet loss [24], Zhang et al. [25], and Lei et al. [26], have structurally adjusted CNN models, optimized CD tasks, and devise various loss functions to mitigate data imbalance issues. Despite these advancements, challenges persist in extracting global spatial–temporal information due to the inherent locality of convolution operations, prompting efforts to integrate attention mechanisms by Fang et al. [27], Chen et al. [23], Liu et al. [28], and Peng et al. [29], although with constraints on global information extraction.

The transformer architecture is renowned for its strong semantic expression capabilities and efficient processing of long-range dependencies. Since the introduction of vision transformer (ViT), transformers have been widely adopted in the computer vision (CV) field to accomplish diverse tasks, including image classification [30], [31], [32], [33], object detection [34], [35], [36], semantic segmentation [32], [34], [37], and image generation [38], [39]. Having achieved

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significant milestones in the CV domain, transformer has also made inroads into the RS field, contributing to tasks such as image time-series classification [40], [41], hyperspectral image classification [42], scene classification [43], and RS image captioning [44], [45]. Concurrently, CD methods have embraced the transformer trend. Esser et al. [39] employed multiple Swin-Transformer blocks to aggregate multilevel feature maps from each image captured by the CNN, subsequently computing different features between bitemporal images. However, this approach incurs substantial resource consumption in terms of memory and computation. Similarly, Zhang et al. [46] devised a pure Swin-Transformer with a Siamese U-shaped structure for the CD task, utilizing an exaggerated 32 Swin-Transformer blocks. In contrast, Chen et al. [47] proposed a bitemporal transformer (BIT) to efficiently model spatial–temporal contexts, comprising a transformer encoder and two transformer decoders in a lightweight manner. Bandara and Patel [48] introduced a Siamese CD network based on the transformer, which enhances the extraction of RS scene features at different scales by integrating features at various levels of the transformer network. Huang et al. [49] utilized the transformer structure to achieve spatiotemporal enhancement through coordinated attention (CA) and concatenated convolution, along with interlayer fusion to improve the extraction capability of RS scene change features. In addition, Chen et al. [50] replaced the transformer structure with the recently proposed Vision Mamba to extract long-term dependencies, yielding promising results. Despite the advancements, optimizing the BIT architecture remains a consideration. For instance, simplifying the transformer encoder to single-head attention could enhance feature coupling and information transmission efficiency. In addition, substituting transformer decoders with lightweight modules may offer further optimizations. While the adoption of the transformer has bolstered model performance, a substantial number of labeled samples are still required for training, akin to CNN-based models.

B. Self-Supervised Learning-Based RS Image CD

Self-supervised learning represents a contemporary paradigm in DL, characterized by its reduced dependency on a vast array of annotated samples and its substantial progress in recent years. Models within the self-supervised learning framework typically comprise a feature extractor and a projector, with training divided into pretraining and downstream fine-tuning phases. During the pretraining phase, pretext tasks are emphasized, with current practices in the CV domain predominantly employing CL [8], [9], [10] and MIM methods [11], [12], [13]. The CL-based method involves image augmentation to generate multiple views, where views originating from the same image are deemed positive samples, while those from distinct images are considered negative samples. These samples serve as training data to facilitate model training. Conversely, the MIM-based method entails applying masking operations to images, utilizing the masked images as samples and the original images as labels to train the model. Both pretext tasks effectively enhance the capabilities of the feature extractor. Subsequently, in the downstream fine-tuning stage, specific decoder and head structures are integrated based on the pretrained feature extractor to enable the model to address targeted tasks. Given that the feature extractor has already acquired the ability to extract common features post-pretraining, only a limited number of labeled samples from downstream tasks are requisite for fine-tuning the model.

Motivated by these advancements, researchers have integrated self-supervised learning techniques into CD tasks, yielding promising outcomes. Tao et al. [51] employed a self-supervised framework to categorize RS scenarios using extensive unlabeled datasets during pretraining. Subsequently, they fine-tuned the well-trained model with a limited number of labeled samples, effectively transferring learned representations to the target task. This method outperformed traditional supervision methods in scenarios with limited labeled data. Saha et al. [52] proposed a self-supervised learning-based CD approach that combines deep clustering, Siamese networks, and CL strategies, achieving satisfactory performance with minimal unlabeled data. Wang et al. [14] utilized a multiview change (MVC) projector and pseudochange augmentation (PeCA) strategy to map change representations into real and pseudo-MVC projections. This approach facilitated the learning of specific discriminative features by segregating predefined positive and negative samples in the projection space. In addition, Wang et al. [15] devised two cascading models, incorporating agent CD learning alongside CL during the pretraining phase to enhance change perception capabilities. Qu et al. [16] leveraged a self-supervised learning framework to extract information from bitemporal images and their differential counterparts, progressively learning potential correlations through an alternating iteration learning strategy. Furthermore, Jiang et al. [53] conducted global and local CL to mitigate the limited local perception issue in RS images, attributed to CL methods. Zhang et al. [17] combined CL with the MIM method, performing postmask reconstruction of RGB images within an online network. The target encoder extracted features from elevation images and computed comparison losses with the online encoder, leveraging the benefits of CL and MIM methods.

Most of the aforementioned self-supervised methods adhere to the CL framework. While Zhang et al. [17] have partially incorporated MIM, it remains primarily grounded in CL principles. CL, which operates based on positive and negative samples akin to clustering methods, primarily trains the global perception capabilities of the model. Changes are inherently more localized in nature, so self-supervised training using CL methods exhibits certain limitations. Some approaches introduce a separate pretext task for capturing change information, which may not only fulfill the training requirements for change perception but also introduce confusion regarding the learning focus, potentially hindering the synergy between these two abilities. In contrast, the MIM method offers a pixel-level training approach, ensuring that the model comprehensively learns both global and local perception abilities. However, as the input for CD tasks comprises bitemporal images, existing MIM methods only reconstruct a single image for training the model. Consequently, this article proposes, for

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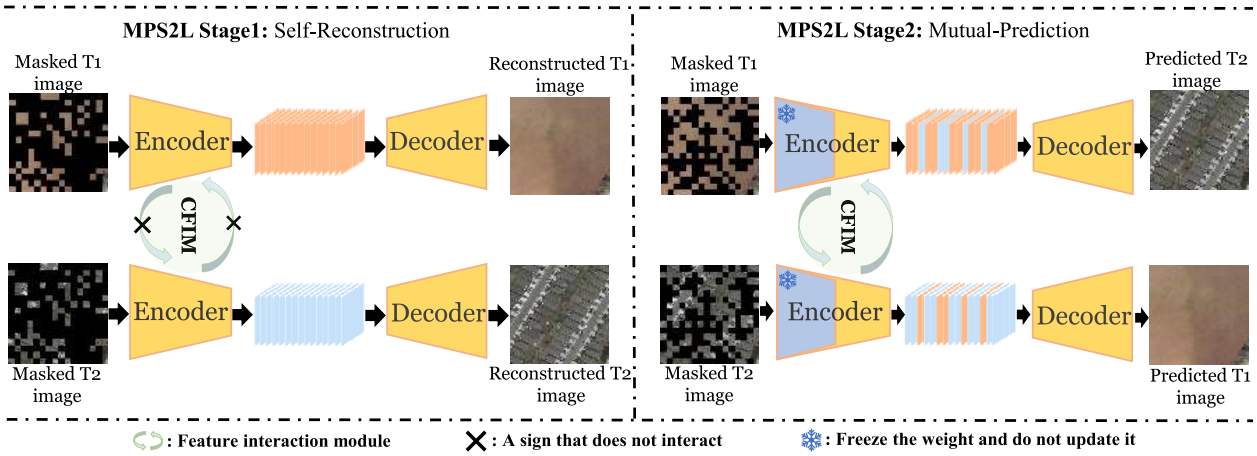


Fig. 2. Schematic overview of the proposed MPS2L method. MPS2L comprises self-reconstruction and mutual prediction stages for self-supervised training. In the self-reconstruction stage, employing the MIM methods, 75% of the patches in the bitemporal images are masked and fed into the encoder to extract common features. Subsequently, these features are passed through the decoder to reconstruct the images, thereby training the ability of the model to perceive global and local RS scenes. During the mutual prediction stage, the CFIM is activated. Initially, 50% of the patches in the images are masked and processed by the encoder to extract common features, followed by capturing change features through the CFIM. Ultimately, the model is directed to the decoder to predict images from another phase, thus honing its capability to capture changes.

the first time, an MIM framework specifically tailored for bitemporal images.

III. Methodology

A. Overview of Framework

The overview of the MPS2L is shown in Fig. 2. Similar to most MIM methods, the proposed MPS2L follows an encoder–decoder structure. However, since the input is bitemporal images, it needs to be adjusted to twin network architecture. This architecture comprises two sets of encoders and decoders with identical structures and parameters. Moreover, the CFIM is incorporated between the twin networks to identify and engage with change information within the features extracted from the bitemporal images by the encoder.

The training of the MPS2L is structured into two stages: the self-reconstruction stage and the mutual prediction stage. During the self-reconstruction stage, the process begins with a masking operation on the image, followed by the input of the masked image into the encoder for feature extraction. Subsequently, these features are fed into the decoder for image reconstruction. This stage does not involve feature interaction, and the parameters of the CFIM remain frozen. In comparison, the decoder's structure is simpler than the encoder, which compels the encoder to develop a more robust feature extraction capability. While the reconstruction of a single masked RS image primarily trains the ability of the model to understand the RS scene, it does not sufficiently enhance the model's capability to perceive change information.

In the mutual prediction stage, the encoder not only extracts features from the current phase masked image but also acquires features from the other phase that are closely associated with changes through CFIM. During this stage, features containing both RS scene and change information are input into the decoder to facilitate the prediction of the original image of another phase. Following training in the self-reconstruction stage, the encoder possesses a sufficiently robust understanding of RS scenes. Subsequent training in the mutual prediction stage further enhances its ability to capture change features, rendering it more adept at handling CD tasks.

Upon concluding the self-supervised training, the performance of the model in the CD task remains underdeveloped. To address this, we preserve the weights of the encoder and the CFIM while substituting the decoder and head with components specifically designed for CD. Subsequently, a small number of labeled samples are utilized for downstream task-specific transfer training. This strategy enhances the adaptability of the model to CD tasks and significantly improves its performance.

B. CFIM

We drew inspiration from Fang et al.'s [18] changer, which demonstrated that feature interaction enhances the ability of the model to perceive change information and promotes similarity in feature distribution between the two branches, facilitating automatic domain adaptation across bitemporal domains. Building upon this insight, we incorporated spatial feature interaction after stage 2 and channel feature interaction after stages 3 and 4. However, in Changer, 50% of spatial dimension elements or channels are randomly selected for the exchange, potentially disrupting feature continuity and leading to the exchange of unrelated features, thereby impeding model training.

To address this issue, we introduce the CFIM based on the attention mechanism, as illustrated in Fig. 3. This approach comprises two submodules: the SIM and the CIM. The SIM computes the spatial attention map $\boldsymbol{M_s} \in \mathbb{R}^{1\times H\times W}$ using the features extracted from bitemporal images and exchanges the corresponding spatial elements based on this attention map. Similarly, the CIM calculates the channel attention vector $\boldsymbol{v_c} \in \mathbb{R}^{C\times 1\times 1}$ from the features of bitemporal images and swaps the channels corresponding to these features according to the

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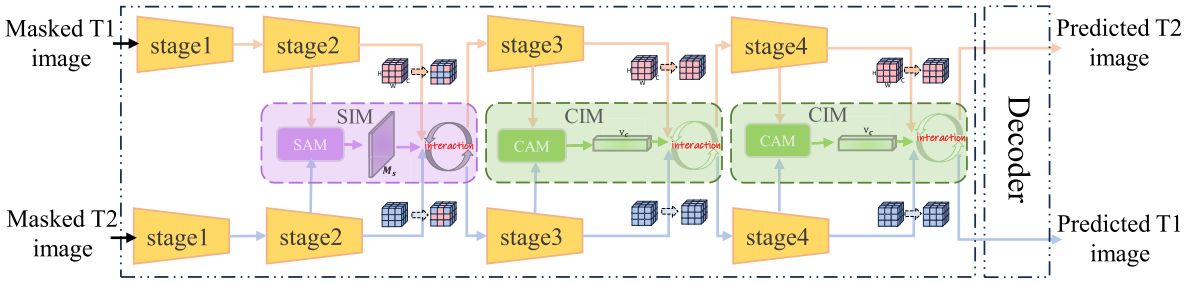


Fig. 3. Overview of the encoder with feature interactions. CFIM conducts spatial feature interaction through the SIM module after stage 2 and channel feature interaction through the CIM module after stage 3 and stage 4.

Algorithm 1 Change Feature Interaction Module

Input: $\boldsymbol{F}_1^l, \boldsymbol{F}_2^l, ia_ratio$. $\boldsymbol{F}_1^l, \boldsymbol{F}_2^l$ are the bitemporal images feature map obtained at the stage l of the encoder, and ia_ratio is the proportion of bitemporal features participating in interactions.
Output: $\boldsymbol{F}_1^{l+1}, \boldsymbol{F}_2^{l+1}$. $\boldsymbol{F}_1^{l+1}, \boldsymbol{F}_2^{l+1}$ are the bitemporal feature map after the interaction of the stage l of the encoder, as input to the stage $l+1$ of the encoder.
Begin:
// Obtain the spatial attention map or channel attention vector
if $l = 2$
  $\boldsymbol{M}_s = SAM(\boldsymbol{F}_1^{l'}, \boldsymbol{F}_2^{l'})$
else if $l = 3, 4$
  $\boldsymbol{v}_c = CAM(\boldsymbol{F}_1^{l'}, \boldsymbol{F}_2^{l'})$
// Calculate the number of pixels or channels to interact
$ia_num = interaction_ratio * sizeof(\boldsymbol{F}_1^{l'})$
// Find the index of the first ia_num feature and interacted with
if $l = 2$
  $ia_index = topk(\boldsymbol{M}_s, ia_num)$
  $\boldsymbol{F}_1^{l+1}, \boldsymbol{F}_2^{l+1} = SIM(\boldsymbol{F}_1^l, \boldsymbol{F}_2^l, ia_index)$
else if $l = 3, 4$
  $ia_index = topk(\boldsymbol{v}_c, ia_num)$
  $\boldsymbol{F}_1^{l+1}, \boldsymbol{F}_2^{l+1} = CIM(\boldsymbol{F}_1^l, \boldsymbol{F}_2^l, ia_index)$
End

channel attention vector

$$\boldsymbol{F}_1^{l+1}, \boldsymbol{F}_2^{l+1} = \mathrm{SIM}\left(\boldsymbol{F}_1^{l'}, \boldsymbol{F}_2^{l'}, \mathrm{SAM}\left(\mathrm{Concat}\left(\boldsymbol{F}_1^{l'}, \boldsymbol{F}_2^{l'}\right)\right)\right)$$
$$\mathrm{SAM}(\cdot) = \sigma\left(f^{7\times7}\left(\left[\mathrm{AvgPool}(\boldsymbol{F}); \mathrm{MaxPool}\left(\boldsymbol{F}'\right)\right]\right)\right) \quad (1)$$
$$\boldsymbol{F}_1^{l+1}, \boldsymbol{F}_2^{l+1} = \mathrm{CIM}\left(\boldsymbol{F}_1^{l'}, \boldsymbol{F}_2^{l'}, \mathrm{CAM}\left(\mathrm{Concat}\left(\boldsymbol{F}_1^{l'}, \boldsymbol{F}_2^{l'}\right)\right)\right)$$
$$\mathrm{CAM}(\cdot) = \sigma\left(\mathrm{MLP}\left(\mathrm{AvgPool}\left(\boldsymbol{F}'\right)\right) + \mathrm{MLP}\left(\mathrm{MaxPool}\left(\boldsymbol{F}'\right)\right)\right) \quad (2)$$

where SAM(·) and CAM(·) are the spatial and channel attention modules, $\boldsymbol{F}_i^l$ represents phase i image features obtained at stage l, and $i \in (1, 2), l \in (1, 2, 3, 4)$, $\boldsymbol{F}_1^l \in \mathbb{R}^{C\times H\times W}$.

C. Self-Supervised and Downstream Transfer Training

1) Self-Reconstruction Training: The self-reconstruction training of MPS2L is depicted on the left of Fig. 2. The twin network structure shares weights, yet there is no interaction between the twin networks at this stage. Features from the masked images are independently extracted and then reconstructed

$$\boldsymbol{X}_i^p = \mathrm{Decoder}\left(\mathrm{Encoder}\left(\boldsymbol{X}_i^m\right)\right) \quad (3)$$

where $\boldsymbol{X}_i^m$ represents the phase i masked image and $\boldsymbol{X}_i^p$ represents the phase i prediction image.

The encoder consists of four stages, as shown in Fig. 3, each of which operates as follows:

$$\boldsymbol{F}^{l+1} = \mathrm{LN}\left(\mathrm{TrasnformerBlock}\left(\mathrm{PatchEmbed}\left(\boldsymbol{F}^l\right)\right)\right) \quad (4)$$

where LN(·) and PatchEmbed(·) represent layer norm and patch embed operations. PatchEmbed(·) uses the transformer block in segformer, which can be a transformer block of any structure.

The decoder consists of three stages, each of which operates as follows:

$$\boldsymbol{F}^{l+1} = \mathrm{LN}\left(\mathrm{TrasnformerBlock}\left(\mathrm{Upsample}\left(\boldsymbol{F}^l\right)\right)\right) \quad (5)$$

where Upsample(·) indicates the up sample operation. The upsample rates are [2, 2, 4].

In the self-reconstruction stage, MSE loss was used to predict the loss of masked patches between the image and the original image

$$L_{\mathrm{sr}} = \frac{\sum_{i,j}\left[\left(\boldsymbol{X}_i^p - \boldsymbol{X}_i\right) \cdot \boldsymbol{T}_{\mathbf{mask}}\right]}{\sum_{i,j} \boldsymbol{T}_{\mathbf{mask}}} \quad (6)$$

where L_{sr} represents the loss from the self-reconstruction stage, $\boldsymbol{X}_i$ is the original image of phase i, and $\boldsymbol{T}_{\mathbf{mask}}$ is the mask template, a matrix with the same size as $\boldsymbol{X}_i$, in which element 1 indicates masking and element 0 indicates no masking.

2) Mutual Prediction Training: The mutual prediction training of the MPS2L method is illustrated in the right section of Fig. 2. During this phase, the encoder component of the twin network engages in feature interaction, with the weights of stages 1 and 2 frozen. This is because these stages primarily extract shallow image features, such as geometric and contour features. The self-reconstruction training stage demonstrates satisfactory performance, while the mutual prediction training phase aims to enhance the capability of the model to extract deep features, particularly change information

$$\boldsymbol{X}_2^p, \boldsymbol{X}_1^p = \mathrm{Decoder}\left(\mathrm{IAEncoder}\left(\boldsymbol{X}_1^m, \boldsymbol{X}_2^m\right)\right) \quad (7)$$

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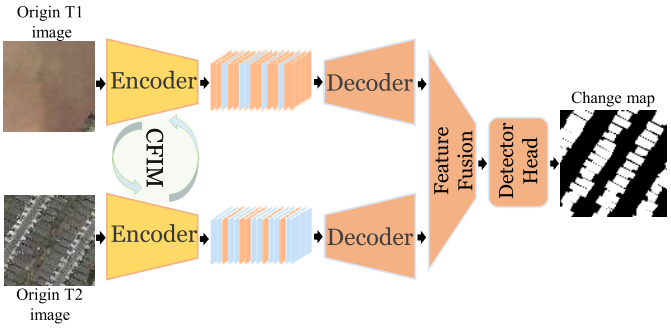


Fig. 4. MPS2L for the CD. The encoder and CFIM and their weights after the self-supervised training are retained, the original decoder is replaced with the decoder designed for the CD task, and a feature fusion module and detector head are also added.

where IAEncoder$(\cdot)$ indicates that a feature interaction policy is added to the encoder.

Then, each stage of operation is

$$\begin{aligned} \boldsymbol{F}_i^{l'} &= \mathrm{LN}\big(\mathrm{TrasnformerBlock}\big(\mathrm{PatchEmbed}\big(\boldsymbol{F}_i^{l}\big)\big)\big) \\ \boldsymbol{F}_1^{l+1}, \boldsymbol{F}_2^{l+1} &= \mathrm{CFIM}[l]\Big(\boldsymbol{F}_1^{l'}, \boldsymbol{F}_2^{l'}\Big). \end{aligned} \tag{8}$$

The decoder in the mutual prediction stage is identical to the self-reconstruction stage, but it employs mean square error (MSE) loss to compute the loss of the predicted image from another phase and the original image in the masked patches' portion

$$\begin{aligned} L_{\mathrm{mp}} = &\frac{\sum_{i,j}\left[\left(\boldsymbol{X}_1^{p} - \boldsymbol{X}_2\right)\cdot \boldsymbol{T}_{\mathbf{mask}}\right]}{2\times \sum_{i,j} \boldsymbol{T}_{\mathbf{mask}}} \\ &+ \frac{\sum_{i,j}\left[\left(\boldsymbol{X}_2^{p} - \boldsymbol{X}_1\right)\cdot \boldsymbol{T}_{\mathbf{mask}}\right]}{2\times \sum_{i,j} \boldsymbol{T}_{\mathbf{mask}}} \end{aligned} \tag{9}$$

where L_{mp} represents the loss from the mutual prediction stage.

3) Transfer Training for Downstream CD: After self-reconstruction and mutual prediction training using bitemporal RS images, the encoder in the MPS2L method acquires a strong understanding of RS scenes and the capability to capture change information. However, self-supervised training primarily focuses on pretext tasks, such as predicting original images, which differs from the CD tasks in downstream applications. Therefore, it becomes necessary to modify the decoder for the CD task and incorporate the projection head based on modifications to the Segformer architecture. In addition, drawing inspiration from the Changer approach, a fusion module has been added to align the bitemporal images, as depicted in the Fig. 4

$$\boldsymbol{Y} = H(\mathrm{Decoder}(\mathrm{IAEncoder}(\boldsymbol{X_1}, \boldsymbol{X_2}))) \tag{10}$$

where $\boldsymbol{Y}$ represents the change map of the final prediction, $H(\cdot)$ represents fusion and project head, and fusion is the FDAF module in the changer. The projection head consists of two convolutional layers.

For the CD downstream task, the conventional cross-entropy is used here to calculate the loss

$$L_{\mathrm{ce}}(\boldsymbol{Y}, \boldsymbol{P}) = -\frac{1}{N}\sum_{i=1}^{N}[\boldsymbol{Y_i}\log_2 \boldsymbol{P_i} + (1-\boldsymbol{Y_i})\log_2(1-\boldsymbol{P_i})] \tag{11}$$

where $L_{\mathrm{ce}}(\boldsymbol{Y}, \boldsymbol{P})$ represents the cross-entropy loss between the CD result $\boldsymbol{P}$ and label $\boldsymbol{Y}$.

IV. Experiments and Results

A. Dataset Description

We utilized three datasets to validate the proposed MPS2L method.

1) *LEVIR-CD [22]:* The LEarning, VIsion, and RS (LEVIR)-CD dataset is a publicly available large-scale building CD dataset. It comprises 637 pairs of high-resolution (0.5 m/pixel) RS images with dimensions of 1024 × 1024. We adhered to its default dataset split for training, validation, and testing. Due to GPU memory constraints, we partitioned the images into nonoverlapping patches of size 256 × 256.
2) *DSIFN [54]:* The deeply supervised image fusion network (DSIFN)-CD dataset is another public binary CD dataset. It encompasses six large pairs of high-resolution (2 m/pixel) satellite images from six major cities in China. This dataset captures changes in various land cover objects, including roads, buildings, croplands, and water bodies. We adopted the default cropped samples of size 512 × 512 provided by the dataset authors.
3) *S2Looking [55]:* The dataset comprises 5000 image pairs (1024 × 1024) and over 65 920 annotated instances of changes derived from side-looking rural area satellite images captured by optical satellites worldwide. These images span a period of one to three years with a resolution ranging from 0.5 to 0.8 m/pixel.

B. Comparison Methods

To verify the effectiveness of the proposed MPS2L method, we compared it with a range of state-of-the-art (SOTA) methods, including supervised and self-supervised approaches.

1) *FC-EF [20]:* A supervised DL CD method that concatenates bitemporal images before passing them through the network. The change map is obtained by conducting "encoded–decoded" operations on the fused image.
2) *CDNet [21]:* This method chains a multisensor fusion SLAM and fast dense 3-D reconstruction pipeline, providing coarsely registered image pairs to a deep deconvolutional network (DN) for pixelwise CD.
3) *SNUNet [27]:* A multiscale feature concatenation method that combines the Siamese network and NestedUNet [55] to extract HR high-level features. Channel attention is applied to the features at each level of the decoder. Deep supervision is also used to enhance the discrimination ability of intermediate features.
4) *BIT [47]:* The method expresses the bitemporal image into a few tokens and uses a transformer encoder to

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model contexts in the compact token-based space–time. The learned context-rich tokens are then fed back to the pixel space for refining the original features via a transformer decoder.

5) *Changeformer [48]:* A CD method based on transformers adopts a twin-structured network mixed with multilayer transformers and MLP, allowing for end-to-end training.
6) *USSFCNet [26]:* A multiscale decoupled convolution (MSDConv) is designed to capture multiscale features of changing objects flexibly by using cyclic multiscale convolution. An efficient spatial–spectral feature cooperation (SSFC) strategy is introduced to obtain richer features.
7) *SEIFNet [49]:* This method captures both global and local information from multilevel features of a Siamese network to enhance the feature extraction capabilities for detecting changes. It incorporates CA, cascaded convolution, and adaptive feature fusion into the decoder to improve CD performance.
8) *MambaBCD [50]:* This method employs the newly proposed Vision Mamba structure as a replacement for the conventional transformer architecture, thereby reducing computational costs while maintaining the ability to capture long-range dependencies.
9) *MAE [8]:* This method uses the MIM approach for self-supervised learning, reconstructing only the unmasked part of the input image.
10) *BYOL [10]:* A comparative learning method without negative samples. The online network has one additional predictor structure, and the target network adopts a momentum update.
11) *SimSiam [9]:* This model uses the twin network structure similar to BYOL and does not utilize negative samples in training.

C. Experimental Settings

The proposed MPS2L was implemented using the PyTorch framework, and the training and testing were conducted on an NVIDIA RTX 4090 GPU. The patch size for the masking operation was set to 8, with a masking rate of 25% during the self-reconstruction phase and 50% during the mutual prediction phase. Throughout all stages of model training, the AdamW optimizer was employed, featuring a momentum of 0.9, a weight decay of $5e^{-4}$, and a learning rate of $5e^{-4}$. The self-reconstruction stage was trained for 1000 epochs, followed by 500 epochs for the mutual prediction stage and an additional 500 epochs for the CD transfer stage.

In the self-reconstruction and mutual prediction stages, the self-supervised model is pretrained using all images of a single dataset. During the transfer training stage for downstream CD tasks, for the LEVIR-CD and DSIFN datasets, 10% of the data region was allocated for supervised training, while 40% of the data region (it was observed that below 40%, none of the methods could be effectively trained) was designated for supervised training of the S2Looking dataset. The remaining samples were reserved for testing purposes. To quantitatively evaluate the methods, this article employs four evaluation metrics: recall, Intersection over Union (IoU), $F1$-score ($F1$), and kappa coefficient (Kappa).

TABLE I
QUANTITATIVE EVALUATION OF DIFFERENT METHODS ON THE LEVIR-CD DATASET

Method	Recall(%)	IoU(%)	F1(%)	Kappa(%)
FC-EF [20]	62.65	52.15	68.55	67.77
CDNet [21]	69.63	56.31	72.05	71.40
BIT [47]	39.20	37.10	54.12	52.67
SNUNet [27]	28.77	27.23	42.81	40.86
Changeformer [48]	60.60	53.98	70.11	69.33
USSFCNet [26]	64.96	59.02	74.23	73.56
SEIFNet [49]	73.19	62.62	78.13	77.42
MambaBCD [50]	69.89	57.02	72.81	73.23
BYOL [10]	27.58	11.14	20.50	17.62
MAE [8]	15.24	14.80	25.78	22.89
SimSiam [9]	19.08	15.32	26.57	24.23
MPS2L(ours)	**75.09**	**67.97**	**80.93**	**80.46**

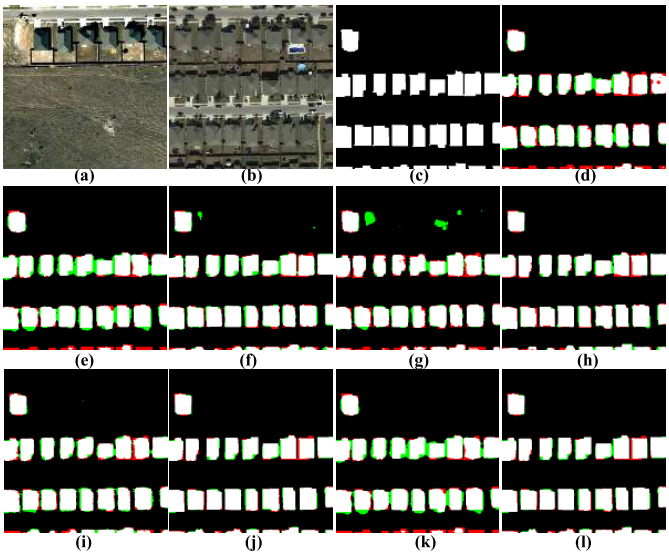


Fig. 5. Visual comparison of the CD maps obtained by different approaches on the LEVIR-CD dataset. (a) T1 image. (b) T2 image. (c) Ground truth. (d) FC-EF. (e) CDNet. (f) BIT. (g) SNUNet. (h) Changeformer. (i) USSFCNet. (j) SEIFNet. (k) MambaBCD. (l) MPS2L. (BYOL, MAE, and SimSiam are the worst of them and not shown.) Different colors are used for a better view, i.e., white for true positive, black for true negative, red for false positive, and green for false negative.

D. Results and Comparisons

1) Experimental Results on LEVIR-CD Dataset: Table I presents the experimental outcomes achieved on the LEVIR-CD dataset, with optimal performance highlighted in bold. It is evident that the performance of the supervised methods diminishes with limited samples but still maintains commendable results. Conversely, self-supervised methods, except for the MPS2L method, exhibit poor performance following training with a small number of samples. Our proposed method attains the best performance in recall (75.09%), IoU (67.97%), $F1$ (80.93%), and Kappa (80.46%). Fig. 5 depicts the CD maps derived from the comparison methods and the proposed MPS2L method. While the supervised methods showcased in the comparison demonstrate commendable results, the self-supervised methods display numerous errors, possibly stemming from inadequate training samples for fine-tuning downstream tasks. However, our proposed MPS2L method

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TABLE II
QUANTITATIVE EVALUATION OF DIFFERENT METHODS ON THE DSIFN DATASET

Method	Recall(%)	IoU(%)	F1(%)	Kappa(%)
FC-EF [20]	79.11	66.04	79.55	68.43
CDNet [21]	81.65	69.42	81.95	72.16
BIT [47]	83.03	74.35	85.29	77.02
SNUNet [27]	82.31	68.83	81.54	71.72
Changeformer [48]	83.33	72.69	84.19	75.52
USSFCNet [26]	88.05	75.06	85.75	78.37
SEIFNet [49]	89.25	79.27	88.44	81.99
MambaBCD [50]	87.22	75.43	83.29	80.87
BYOL [10]	74.94	62.43	76.87	63.90
MAE [8]	90.21	76.99	87.00	80.36
SimSiam [9]	73.86	65.29	79.00	66.41
MPS2L(ours)	**91.04**	**81.58**	**89.85**	**84.49**

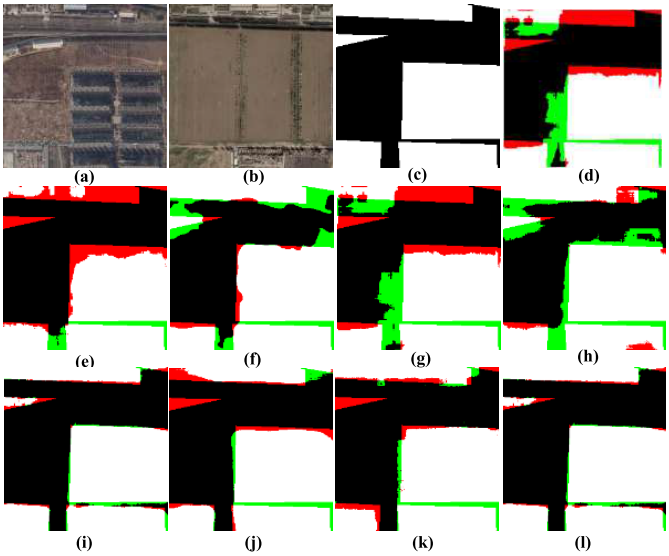


Fig. 6. Visual comparison of the CD maps obtained by different approaches on the DSIFN dataset. (a) T1 image. (b) T2 image. (c) Ground truth. (d) CDNet. (e) BIT. (f) SNUNet. (g) Changeformer. (h) USSFCNet. (i) SEIFNet. (j) MambaBCD. (k) MAE. (l) MPS2L. (FC-EF, BYOL, and SimSiam are the worst of them and not shown.) Different colors are used for a better view, i.e., white for true positive, black for true negative, red for false positive, and green for false negative.

yields superior results compared to the supervised methods. This underscores the efficacy of the mutual prediction mechanism in enhancing the ability of the model to capture the change information.

2) Experimental Results on DSIFN Dataset: Table II presents the experimental results obtained on the DSIFN dataset, with the best performance highlighted in bold. All methods exhibit excellent performance due to the larger area and distinct variation characteristics of the DSIFN dataset. Compared to SEIFNet, the leading supervised method, MPS2L shows notable improvements of 1.79%, 2.31%, 1.41%, and 2.5% in the recall, IoU, $F1$, and Kappa, respectively. In addition, apart from the MPS2L, the MAE methods also demonstrate superior performance compared to the supervised methods, possibly due to the dataset's wider change region, which facilitates easier capture of change information by the model. Compared to the LEVIR-CD dataset, the DSIFN dataset presents a wider variation region, resulting in generally improved model performance after training on this dataset. As depicted in Fig. 6, it is evident that all methods demonstrate good performance, with some self-supervised methods even outperforming certain supervised methods. Notably, our proposed MPS2L exhibits the most effective results among them.

TABLE III
QUANTITATIVE EVALUATION OF DIFFERENT METHODS ON THE S2LOOKING DATASET

Method	Recall(%)	IoU(%)	F1(%)	Kappa(%)
FC-EF [20]	60.07	25.63	40.80	40.29
CDNet [21]	65.75	30.67	46.94	46.47
BIT [47]	75.83	40.63	57.78	57.38
SNUNet [27]	**76.72**	36.75	53.75	53.34
Changeformer [48]	75.52	37.56	54.61	54.20
USSFCNet [26]	75.72	34.73	51.55	51.14
SEIFNet [49]	72.59	46.64	63.61	63.20
MambaBCD [50]	73.62	44.26	60.88	60.11
BYOL [10]	68.22	38.30	55.38	54.93
MAE [8]	73.73	32.82	49.42	48.99
SimSiam [9]	72.71	37.62	54.68	54.25
MPS2L(ours)	72.66	**47.05**	**63.99**	**63.59**

TABLE IV
EFFICIENCY COMPARISON ANALYSIS OF DIFFERENT CD METHODS

Method	FLOPs(G)	Params(M)
FC-EF [20]	3.57	1.35
CDNet [21]	23.47	1.43
BIT [47]	15.17	17.99
SNUNet [27]	54.83	12.03
Changeformer [48]	21.18	29.75
USSFCNet [26]	4.86	1.52
SEIFNet [49]	10.8	12.71
MambaBCD [50]	69.18	79.24
BYOL [10]	37.07	17.69
MAE [8]	59.57	19.74
SimSiam [9]	8.37	27.91
MPS2L(ours)	11.55	21.11

3) Experimental Results on S2Looking Dataset: Table III presents the quantitative evaluation results. Except for recall, the MPS2L method has the highest index on the S2Looking dataset, with IoU, $F1$, and Kappa values of 47.05%, 63.99%, and 63.59%, respectively. The method with the highest recall value is SNUNet, which is 76.72%. Other self-supervised methods and supervised methods exhibit similar performance, potentially being attributed to the complexity of the variation region within the dataset. The S2Looking dataset encompasses a broader expanse compared to the LEVIR-CD and DSIFN datasets, yet it exhibits higher noise levels and more intricate variations. As depicted in Fig. 7, it is evident that a majority of the supervised and self-supervised methods manifest high error rates on this dataset.

E. Model Efficiency

To evaluate model efficiency, we conducted an analysis of all comparison algorithms in terms of the number of floating-point operations (FLOPs) and parameters (Params). FLOPs and Params were used to measure the computational and spatial complexities of the models, respectively. Table IV presents these metrics for input images of size 256 × 256 pixels. As indicated in Table IV, the FC-EF method exhibits the lowest FLOPs and Params while main-

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TABLE V
ABLATION STUDY OF INTERACTION STRATEGY ON LEVIR-CD AND S2LOOKING DATASETS. IOU, $F1$, AND KAPPA ARE REPORTED

Interaction Strategy	LEVIR-CD			S2Looking		
	IoU(%)	F1(%)	Kappa(%)	IoU(%)	F1(%)	Kappa(%)
no interaction	51.45	67.45	66.77	33.73	50.47	50.14
random interaction	53.39	70.04	69.02	37.22	54.26	54.10
attention interaction	56.70	72.21	71.52	40.10	56.86	56.38

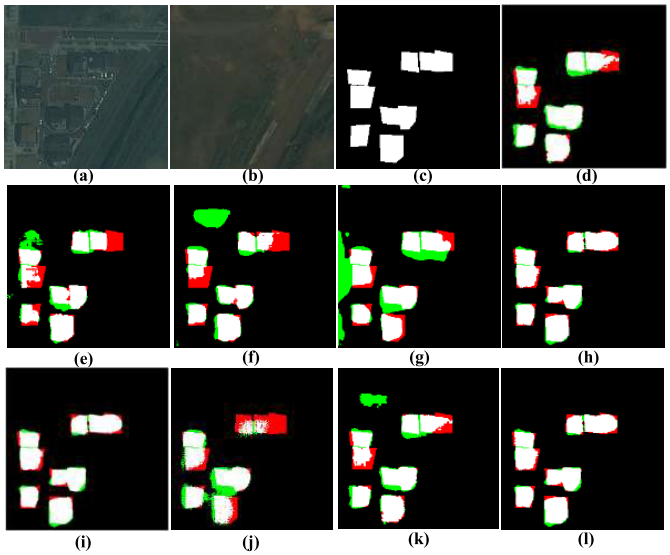


Fig. 7. Visual comparison of the CD maps obtained by different approaches on the S2Looking dataset. (a) T1 image. (b) T2 image. (c) Ground truth. (d) BIT. (e) SNUNet. (f) Changeformer. (g) USSFCNet. (h) SEIFNet. (i) MambaBCD. (j) BYOL. (k) SimSiam. (l) MPS2L. (FC-EF, CDNet, and MAE are the worst of them and not shown.) Different colors are used for a better view, i.e., white for true positive, black for true negative, red for false positive, and green for false negative.

taining good performance across the three datasets. USSFCNet shows a slight increase in FLOPs and Params compared to the FC-EF method but achieves significantly improved performance, surpassing most methods with higher FLOPs and Params. The MAE method, despite having the highest FLOPs and Params, performs comparably to many supervised methods specifically designed for CD tasks. Changeformer exhibits the highest FLOPs and Params among the CD methods, yet its performance does not surpass that of other supervised methods, possibly due to the limited training data. Our proposed MPS2L method demonstrates higher Params due to the utilization of the transformer structure but achieves lower FLOPs as a result of our model's optimized data processing.

F. Ablation Studies

1) Ablation on CFIM: The CFIM was analyzed through ablation experiments to evaluate its efficacy in exchanging change-related features during the model encoding phase. Three experimental sets were conducted: 1) with frozen feature interaction module parameters, thus no interaction; 2) employing a random 50% feature interaction strategy; and 3) utilizing the attention-based interaction strategy that we designed (see Table V).

In Table V, it can be seen that the performance of the model with random 50% feature interaction is improved compared with that without feature interaction, and the $F1$ is increased by 2.59% (LEVIR-CD)/3.79% (S2Looking). The performance of feature interaction based on attention mechanism is also improved compared with random 50% feature interaction, in which $F1$ is increased by 2.17%/2.6%. We believe that the performance is improved because the attention mechanism enables the feature interaction module to improve its ability to capture change information. Previously, the random 50% interaction strategy required a backbone to capture change information.

2) Ablation on Mutual Prediction Mechanism: We conducted an ablation study on the mutual prediction mechanism to evaluate its efficacy. Four experimental groups were conducted: 1) without self-supervised learning; 2) only masked image reconstruction training; 3) only masked image mutual prediction training; and 4) with early reconstruction training followed by late mutual prediction training.

As can be seen in Table VI, the performance of the model without self-supervised training is the worst, which may be due to the use of transformer structure as encoder and decoder, which has poor performance when samples are small. The performance of the model improved only after self-reconstruction training. Compared with no self-supervised training, $F1$ increased by 4.61% (LEVIR-CD)/3.06% (S2Looking). Compared with only self-reconstruction training, the performance of the model improved more after mutual prediction training, and the $F1$ increased by 1.22%/2.33%. It can be seen that mutual prediction is more suitable for the CD task. After the model underwent self-reconstruction training and then mutual prediction training, $F1$ increased by 2.89%/1.74% again compared with only mutual prediction training.

V. DISCUSSION

A. Feasibility of Mutual Prediction

To gain a deeper understanding of the capability of MPS2L to achieve cross-temporal mutual prediction between different phases using partial patches and change features from RS images, we selected a pair of bitemporal RS images from the LEVIR-CD, DSIFN, and S2Looking datasets. These images underwent a 50% masking operation before being input into the fully trained MPS2L for mutual prediction operations. As illustrated in Fig. 8, MPS2L has demonstrated a favorable

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TABLE VI
ABLATION STUDY OF MUTUAL PREDICTION MECHANISM ON LEVIR-CD AND S2LOOKING DATASETS. IOU, $F1$, AND KAPPA ARE REPORTED

Self Reconstruction	Mutual Prediction	LEVIR-CD IoU(%)	LEVIR-CD F1(%)	LEVIR-CD Kappa(%)	S2Looking IoU(%)	S2Looking F1(%)	S2Looking Kappa(%)
✘	✘	56.70	72.21	71.52	40.10	56.86	56.38
✔	✘	59.89	76.82	76.54	43.12	59.92	59.77
✘	✔	62.39	78.04	77.89	45.71	62.25	62.02
✔	✔	67.97	80.93	80.46	47.05	63.99	63.59

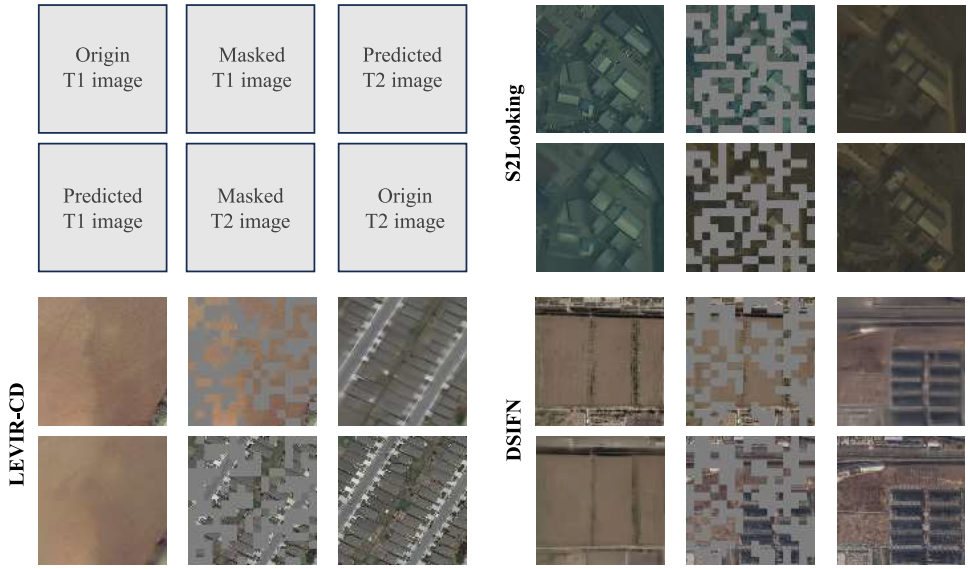


Fig. 8. Visualization of mutual prediction results. The bitemporal RS image location layout of the three data sets is referred to in the upper left.

predictive effect for the samples' postmask operation across the three datasets. It accurately predicted both the overall color changes attributed to seasonal variations and the local changes resulting from human activity, albeit with some minor ambiguity.

B. Influence of Sample Size for Transfer Training

Other self-supervised methods, aside from MPS2L, exhibit poor performance when transferring to downstream tasks using only 10% of the LEVIR-CD dataset for training. However, upon training with 10% of the DSIFN dataset and 40% of the S2Looking dataset for downstream task transfer, these methods achieve similar or even superior results compared to supervised training. To investigate the influence of labeled sample size on downstream task migration performance, experiments were conducted using 30%, 50%, and 70% regions of the LEVIR-CD dataset for training, with the remaining regions designated for testing. Comparative experiments were performed on SEIFNet (representing the SOTA in supervised methods), MPS2L, BYOL, SimSiam, and MAE, with the $F1$ score serving as the reference evaluation metric.

The experimental results are shown in Fig. 9. The results indicate that increasing the labeled samples from 10% to 30% regions enhances the performance of all self-supervised methods. However, the performance of self-supervised methods, with the exception of MPS2L, remains inferior to SEIFNet

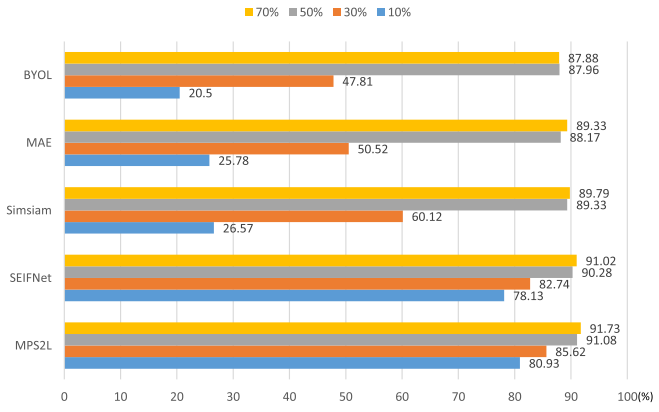


Fig. 9. $F1$-score of the selected method on different transfer training sample sizes.

when using only 10% of the data. Increasing the proportion of labeled samples from 30% to 50% significantly enhances the performance of all methods although the self-supervised methods, other than MPS2L, continue to underperform compared to SEIFNet. Further increasing the labeled samples from 50% to 70% results in limited performance improvement across all models, with the performance of the BYOL method even slightly declining, possibly due to errors introduced by the random selection of training sample areas.

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These results highlight the ongoing necessity of a certain number of labeled samples for downstream task transfer training following self-supervised training. In addition, a more optimized model design could potentially reduce the reliance on labeled samples for transfer training.

VI. Conclusion

In this article, we propose the MPS2L tailored for the RS image CD task. Compared with the CL methods, the pixel-level self-supervised learning strategy superiorly enhances the ability of the model to interpret the local region of the RS scene and facilitate the detection of small-scale objects. CFIM carries out spatial interaction through shallow feature computing spatial attention map to promote the extraction of local change features and carries out channel interaction through deep feature computing channel attention vector to promote the extraction of global change features. This cross-temporal mutual prediction mechanism makes the training of the ability of the model to understand RS scenes and its ability to capture change information belong to the same optimization goal of "prediction," avoiding the existence of multiple optimization goals that lead to the model not being effectively or adequately trained. At the same time, it also helps to enhance the ability of the model to recognize change information and generalization ability in subsequent transfer training. Extensive experimental evaluations confirm the efficacy of the mutual prediction mechanism and the attention-based feature interaction module. Comparative analysis with existing self-supervised and supervised methods shows that the MPS2L method has superior performance on popular data sets and demonstrates strong detection capability in identifying areas of change.

Due to domain differences between various datasets, the MPS2L method in this experiment utilizes RS images from only a single dataset during the self-supervised training stage, rather than images from all datasets. Incorporating a larger number of samples during self-supervised training can enhance the feature extraction and generalization abilities of the model. Consequently, the limited self-supervised training samples used in this experiment may have constrained the performance improvement of the MPS2L method to some extent. In future work, we plan to develop a strategy to address domain differences when using different datasets in the self-supervised training phase, enabling the MPS2L model to be trained on a sufficient number of RS images, thereby further improving its performance.

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